

Let's Test 2017

Technical Web Testing and Automating From the Web Developer Tools

Alan Richardson

"a combination of demonstrations, hands on exercises, and challenges to interact with, test and automate simple web applications, all from the browser"

- www.eviltester.com
- twitter: @eviltester

Blurb - What exactly is a “Testing Game”?

Some people would have you believe it is an off the shelf card game involving wiggly lines, or a ‘fun interactive’ conversation game telling stories, or some other ‘game mechanism’ related to logical reasoning. Or perhaps a party game with dice, optional blindfolds, and random plastic paraphernalia, but never knives. Other people might show you a board game or two, perhaps for 1-4 players of ages 12+ or any age but with adult supervision, two or more dice and possibly involving kittens.

And all of those people would be WRONG.

Blurb - A “Testing Game” is...

A “Testing Game” is a game designed and programmed by one or more testers for the purposes of playing by testers. Sure you’ll have fun, but it must also frustrate the player (tester) with an outrageous bug at the most inopportune moment, or an unplayable game without a GUI so you have to use tools to play it, or a game with ‘much’ of the game play exposed, but with the high point scoring opportunities open only to those who have been inducted into the secrets of testing... the secrets that will be shown in this session.

Blurb - If you are brave enough...

If you are brave enough to enter the “Evil Testers’ Arena of Testing Games of Evil Testing” then you shall learn the secrets of:

- Viewing Source (Yes! The Ultimate Skill!)
- Browser Developer Tools
- Debugging JavaScript
- Using the JavaScript Console
- Testing REST Services from the Browser
- REST API Clients
- Custom Headers
- PUT, POST, GET Requests
- The secret of the voodoo drum in the zombie graveyard

Blurb - More Secrets

And more secrets that will only remain secret until the secrets are revealed in this 'open to anyone of any skill level so long as they can breathe and have a laptop' session (feel free to use a tablet if you want to get beaten by everyone else because you'll be vulnerable without tools, unless you can find equivalent tools for a tablet in which case you might be able to use your skills and knowledge to win, but probably not, because laptopz roole!)

- Do you have a laptop?
- Can you connect to the internet?

Blurb - You will play:

- Games of dexterity and skill that run in a browser
- Games with obvious bugs
- Games with devious, secret bugs
- Games with no bugs that we know of but that are just hard to play unless you have the technical skills to do so
- Games with no bugs that we know of but probably have bugs because they are games implemented as software on a computer that you can test
- RestMud with new adventure maps and functionality
- Multi-player RestMud – yes, with other real people, in a virtual dungeon, really doing virtual things

Enter “The Evil Tester’s Arcade of Testing Games of Evil Testing” and:

- play
- find bugs
- fix bugs
- write code
- fix code (We have bots that play games badly – you could make them play better)
- develop new skillz
- use the skillz you have to earn killz

This is not a workshop about gameification. This is workshop about games for edification and evisceration.

Are you game?

Mix of Skills Supported

- I assume no knowledge
- If the content is new to you then follow the demos and examples
- If you already know a section then use the 'games list' to experiment
- Please be considerate of others:
 - in a demo/lecture/debrief section no talking to neighbour,
 - do engage with the group
 - if you know something, share it with the group
 - if you learned something, share it with the group
 - Questions? Ask - someone else may want to know too

Content Overview

- learn how to tactically automate from Dev Tools console with JavaScript
- learn how to observe, interrogate and manipulate the DOM
- learn basics of JavaScript
- learn basics of Dev Tools

Slides and Handouts and Links

<http://compendiumdev.co.uk/page/atLetsTest2017>

- Also on the bottom of every slide.

An Intro Demo

Z-Type

- Z-Type online zty.pe
- A typing Game
- Bot is advanced
 - but at the end of this workshop you should understand it
 - you might need more practice to be able to write it

A Z-Type Bot

```
var zTypeBotAlphaChars = "abcdefghijklmnopqrstuvwxyz";  
var zTypeBotCharToType = 0;  
ztypebot = setInterval(function() {  
    ig.game.shoot(  
        zTypeBotAlphaChars.charAt(zTypeBotCharToType));  
    zTypeBotCharToType = (zTypeBotCharToType + 1) % 26;  
    }  
    , 100);
```

Above bot is very crude but can play the game.

```
clearInterval(ztypebot);
```

A Quick Demo of a Real App

- Because this isn't just games
- todomvc.com/examples/backbone
- Demo to show how automating from the console supports us as a tactic for interactive and exploratory testing

Create Some Data in the App

```
for (x = 0; x < 100; x++) {  
  app.todos.create({  
    title: "todo ".concat(x),  
    order: app.todos.nextOrder(),  
    completed: false  
  })  
}
```

Delete Them All

```
for (x = app.todos.models.length - 1; x >= 0; x--) {  
  app.todos.models[x].destroy()  
}
```

Demo Section

Hands off keyboards

- Slide based demo section to provide an overview

Explanation and Demo of Basic Concepts

- We will have a hands on section after this
- Do ask questions as we go through

You Will See

- Use of Browser Dev Tools
- HTML & DOM Interaction
- JavaScript Console Interaction
- Basics of JavaScript

After this, we will have a hands on guided walkthrough where you follow along.

About the Demo App

- The Evil Tester Sloganizer
 - <http://www.eviltester.com/apps/sloganizer/version/1/sloganizer.html>
- Single Page App
- Generates Random "Evil Tester" Slogans
- Periodically shows an 'ad' for Book "Dear Evil Tester"

Every Browser Allows Us To

- View HTML Source
- Manipulate DOM
- Execute JavaScript

View Source

- Raw, Unformatted
- The code sent from the server
- Might have syntax highlighting
- Might show syntax errors

'Inspect' to View the DOM

- View the DOM
 - Document Object Model
 - How the HTML is interpreted
 - See Styling

'Inspect' to Amend the DOM

- Anything in the browser can be amended
 - e.g. Change button text
- Amend Applied CSS styles
 - e.g. "display:none"
- Edit HTML Blocks
 - e.g. amend the ad

Other Dev Tools

- 'Console' view JavaScript errors, debug statements and execute JavaScript
- 'Sources' view all pages and source files used
 - set breakpoints for JavaScript debugger
- 'Network' view Network traffic
- etc. Cookies, local storage

JavaScript Console

- Any JavaScript Errors shown here
- Debug Messages shown here
- Can also execute arbitrary JavaScript
 - `changeSlogan()`

HTML and JavaScript

- Inspect button

```
onclick="changeSlogan()"
```

- see also `Event Listeners`
- see also `<script>` sections in page
- see also any `.js` files in 'Sources' and 'Network'

changeSlogan

- call with `()`
- view without `()`

JavaScript Basics

- Objects
 - 'things' with properties and functions
- Functions
 - things you can 'call' to do stuff
 - calls other functions
 - return values to calling functions
- Variables
 - things with values you can change
 - `var myName = "Alan";`

Objects in Sloganizer

- an object called `phrases` which is a set of phrases, and these 'phrases' are arrays
- an object called `sentences` which is an array of `string`

Both of these objects (`phrases` , `sentences`) are assessible from the JavaScript console

If I type `phrases` at the console I will see information about that object in the console.

Functions

Functions look a bit like this:

```
function randomNumberUpTo(theVal){  
    return Math.floor(Math.random() * theVal)  
}
```

- This is a `function`
- called `randomNumberUpTo`
- it takes a single argument `theVal`
- it uses the JavaScript `Math` library
- it returns a `random` number up to the value of `theVal`

Some Functions in Sloganizer

- `random_sentence()`
- `process_sentence(sentence)`

Could call `random_sentence()` directly, need to pass an argument into `process_sentence`

```
process_sentence(sentences[0])
```

Arrays

The `sentences` object is an 'array'.

- `sentences[0]` would return the first item in the array
- `sentences.length` would return the length of the array
- `sentences[sentences.length-1]` would return the last value in the array

Can change values in arrays:

- `sentences[0] = "bob"` - change the first entry in the array to "bob"

For Loops

```
for(var x=1;x < 10; x++){  
    console.log(sentences[x]);  
}
```

- start x at 1
- while x is less than 10
- write the sentences array content at index x to the console
 console.log
- increment x (x++ means add 1 to x)

Advanced Techniques

setInterval

```
var sloganbot = setInterval(function(){  
    changeSlogan()  
},1000)
```

create a thread in variable `sloganbot` that calls `changeSlogan` every `1000` milliseconds.

Stop it with `clearInterval`

```
clearInterval(sloganbot)
```

Summary

- HTML and DOM can be different
- Use Dev Tools to View and Amend DOM
- Dev Tools let you view all sources loaded into browser
- Dev Tools Console
 - lets you interact with JavaScript Interpreter
 - write code to interact with application

JavaScript Summary

- Objects, Functions, Arrays, Variables
 - all accessible and createable from console
- for loops increment a variable until a condition is met
 - `for(var x=1;x < 10; x++){console.log(x);}`
- call functions with `()` - add any arguments within `()`

JavaScript 'Bots'

- `setInterval` to execute a function every X milliseconds (`1000`)

```
var mybot = setInterval(function(){  
    /* add the code you want in here */  
},1000)
```

- store your 'bot' interval in a variable so that you can stop it with

```
clearInterval(mybot);
```

JavaScript Sundries

- `var` to create a variable
- `[0]` access array values with `[]` first item starts at `0`
- end statements with `;`
- comments `/* span lines or in lines */` and `// comment to end of line`

The Handouts contain a 'Just Enough JavaScript To Be Dangerous' section with more details of all of this.

Walkthrough Section

Hands on keyboards

Play Along At Home

- copy and paste from the document if you need to
- you can't break anything
- ask questions if it isn't clear

Visit the Sloganizer site

- The Evil Tester Sloganizer
 - <http://www.eviltester.com/apps/sloganizer/version/1/sloganizer.html>
- or follow link from the workshop page

Understand the Functionality

- click the [I Want A Slogan] button a few times

Understand the HTML

- right click and view source
- basics of HTML
- sections: `head` , `body` , `style` , `script`
- tags: `h1` , `h2` , `ul` , `li` , `div`
- attributes: `class` , `href` , `onclick` , `id` , `style`

Why?

- learn HTML
- review for data leaks
- HTML comments might give more information about app
- some source views show HTML syntax errors

Inspect DOM with Dev tools

- highlights position on page
- expand/collapse DOM elements
- see page structure
- see styles in side menu

Why?

- understand structure of app
- review for data leaks
- review for accessibility
- learn HTML
- view CSS errors and issues

Manipulate DOM with Dev tools

- change text on the button
- find the `bookinfo`
- change class on the bookinfo
- make bookinfo visible by changing style
- delete the `bookinfo`

Why?

- put app in different states
- amend values in forms to send data to server that might not be allowed

View JavaScript Console

- JavaScript errors now visible
- refresh page
- call the `onclick` function `changeSlogan()` from console
 - tab completion,
 - up and down to show command history

Why?

- errors and log messages are visible
- use `console.log` to log values from the app
- explore the app objects and data

Use Sources to view JavaScript

- Chrome (Sources), Edge/Firefox (Debugger), Safari (Resources)
- You could set a breakpoint here
 - only mentioned for more advanced participants - not necessary for the exercises in this workshop

Why?

- Understand the application in more detail
- Test at a lower level
- Learn how to manipulate the app to support testing
- Learn JavaScript

Use Network to view traffic

- Not necessary for exercises

Why?

- see what code is available
- understand the 'app'
- any data leak issues?

Explore the Objects from console

- type `sentences`
- should see it is an array
- type `sentences[1]` gets the second item in the array
- type `sentences.length` (property not a function call)
- type `sentences[sentences.length-1]` (get the last item in the array)

View the source in

Sources/Debugger/Resources

- look at the `changeSlogan` function code
- it calls `random_sentence` to get a sentence
- call `random_sentence` from console
 - type `random_sentence()`

Wrap call to `random_sentence` in a `for` loop

```
for(var s=0; s < 10; s++){  
    random_sentence();  
}
```

How many sentences did you see?

Only 1, but we want to see 10

- `random_sentence()` was called 10 times but only the last value was returned
- we actually want to `console.log` the output from each call

```
for(var s=0; s < 10; s++){  
    console.log(random_sentence());  
}
```

Tip: create a text file if it helps but you lose code completion, in Chrome use Snippets in Sources (right click run)(quick demo)

Why Run code from console?

- test smaller isolated parts of the application
- setup data
- put application in a specific state for testing

Automate the App from the console

```
var sloganbot = setInterval(function(){  
    changeSlogan()  
}, 1000)
```

create a thread in variable `sloganbot` that calls `changeSlogan` every `1000` milliseconds.

Stop it with `clearInterval`

```
clearInterval(sloganbot)
```

Why?

- longer running test
- might expose JavaScript errors that appear over time
- any memory leaks?
- does it randomly generate slogans?
- does the 'ad' appear?
- Observe other parts of the app while it is being used
 - is there any network traffic?
 - monitor local storage and cookies?

Any Questions?

About the games and challenges

The next sections are:

- overview of game or challenge
- play/automate game or take a challenge
- hands on experimentation, I am here to help
- group debrief and Q&A
- some sections have demos and summaries

Games and Challenges

Designed to give you the chance to explore:

- Dev Tools
- DOM Manipulation
- JavaScript Console
- JavaScript Techniques

This is your workshop - Ask Questions - Experiment

If you are not ready to move on yet

- continue working through the challenges and tasks
- read the slides: learn JavaScript - try out the code examples

If current challenge is too basic for you

- solve it then move on to other challenges and games on the list of games or slides
<http://compendiumdev.co.uk/games/buggygames/>
- read the slides, experiment with JavaScript you don't know
- **continue to join in with debriefs, demos, and Q&A sessions**

Console Driver

- a Basic Game
- 5 mins play/test/explore it session
- debrief

Console Driver

The Ultimate Driving Game Experience. You are a road warrior hell bent on speeding along the ultimate winding road. Can you survive longer than anyone else?

- Thrill to the ever increasing speed of travel
- Gasp as you narrowly avoid hitting the lethal road wall
- Put your driving skills to the test with trigger fast controls

Don't Die.

Play Console Driver from
compendiumdev.co.uk/games/buggygames

Console Driver Hints & Suggestions

Hints

- Use the Browser Dev Tools Console
- Have Focus on the Page to play the Game

Suggestions

- Are there any bugs?
- Have you looked at the code?
- Can you cheat?
 - can you score more than "291551004004002940"?

Play Console Driver from

compendiumdev.co.uk/games/buggygames

Console Driver - Debrief

- Knowledge Sharing
- What did you do?
- How did you do it?
- What problems did you find in the game?
- What problems did you have testing it?
- Did you keep notes?
- What tools did you use?

Console Driver - Some *Obvious* Secrets

- infinite game bugs
 - go right very quickly when there are still gaps in wall for infinite game
 - go left very quickly when there are still gaps in wall for invisible infinite game
- cannot amend `points` but can amend `distanceTravelled` which is the basis for `points`
- did you find and try the `setLevel` GUI?
 - did you `promptForValues=true` ?
 - Hint - press ENTER key before starting game.

Console Driver - Look at the source

- `document.onkeydown` is important event handler
- use the source to learn how to write code that manipulates the game e.g. cut and paste out from source into console, change variables, etc.

Console Driver - Hack Targets

- `distanceTravelled`
 - controls the score which is displayed as `points`
- `widthOfTrack`
 - controls the width of the track

All of these could be amended at the console.

e.g. `variableName = value`

Console Driver - Console Hack Targets

Just Enough JavaScript To Be Dangerous

Use the Console to amend variables

e.g.

- `widthOfTrack=30`
- `distanceTravelled+=100` //(add 100 to `distanceTravelled`)

Just Enough JavaScript To Be Dangerous

- Bots

- To 'Create' A Bot that runs every 100 milliseconds

```
var myBot = setInterval(function() {  
    //whatever JavaScript Code I want in here;  
}, 100);
```

- To 'Kill' the Bot

```
clearInterval(myBot);
```

Just Enough JavaScript To Be Dangerous

- **for** , **if** , **array** , **break** , **** , **++** , **==**

```
for (i = 0; i < anArray.length; i++) {  
    if(anArray[i]==" "){  
        // do something;  
        break;  
    }  
}
```

Just Enough JavaScript To Be Dangerous

- `for` loop executes code within `{ }` until a condition
- `for( setup ; until true ; at end of each iteration )`
- arrays have numeric keys and start at `0`
 - e.g. `anArray[0]` is first entry in an Array
- `i++` increment value of variable `i` by `1`
- arrays have a 'length'
- `break` exit a loop early
- `if (condition is true){ then do stuff in here }`

Console Driver Bot Easy Suggestions

Can you write the following Bots?

- Point Increaser Bot
 - every 500 milliseconds increase `distanceTravelled` for a subtle points boost
- Width Of Track Management Bot
 - every 500 milliseconds the bot maintains the width of the track for you

Combine Bots and/or Cheats for Mega Points

Console Driver Bot Challenges

Can you write more advanced Bots?

- Speed Brake Bot
 - every 500 milliseconds the Bot sets the speed to a manageable speed for you
- Create a Bot that plays the game
 - No hints - look at the code and figure it out

Point Increaser Bot - Console Driver

```
var myPointsBot = setInterval(function() {  
    distanceTravelled=distanceTravelled+50;  
}, 500);
```

and

```
clearInterval(myPointsBot);
```

Width Of Track Management Bot

```
var myWidthBot = setInterval(function() {  
    widthOfTrack=58;  
}, 500);
```

and

```
clearInterval(myWidthBot);
```

Speed Bot - Console Driver

Cannot just change `millisecondsBetweenScrolls`, need to amend difficulty levels.

- set to lowest level (0) for slow speed, or fastest level (6)

```
var mySpeedBot = setInterval(function() {  
  setDifficultyLevel(0);  
}, 500);
```

and

```
clearInterval(mySpeedBot);
```

A Game Play Bot - Console Driver

```
var myCarControlBot = setInterval(function() {  
  // using code from `displayTheTrackToTheConsole`  
  var trackDisplay = getTrackLine(posOfLeftOfTrack,  
                                   widthOfTrack, maxWidthOfArena);  
  var trackArray = trackDisplay.split("");  
  var carSpace = trackArray[carPosition];  
  for (i = 0; i < trackArray.length; i++) {  
    if(trackArray[i]==" " && trackArray[i+1]==" "){  
      carPosition=i+1;  
      break;  
    }  
  }  
}, 50);
```

```
clearInterval(myCarControlBot);
```

Questions?

The Coloured Square Game

Click the link to change the colour, when it is the colour you want then press the button.

The Ultimate Game of colour matching and cheating.

How many ways can you cheat to win?

- compendiumdev.co.uk/games/buggygames
 - [/the_coloured_square_game/colouredsquare.html](http://compendiumdev.co.uk/games/buggygames/the_coloured_square_game/colouredsquare.html)

Play The Game

- Play The Game
- Look at the Code
- Any Bugs?

5 Minutes

Debrief

- What did we learn?

Known Bugs

- square starts as grey but never turns back to grey
- it never turns "red"
 - "red" is not in the array of colours

```
var colours = ["blue", "white", "green", "black", "pink"];
```

Quick Array Overview

- View array at console
 - `colours`
- create an empty array
 - `colours = [];`
- set a value in array
 - `colours[0] = "green";`

DOM interaction with JavaScript

Overview

- find an element by id
 - `document.getElementById("changeLink")`
- get styles etc.
 - `element.style.backgroundColor`
- can change styles etc.
 - `= .style.backgroundColor = "black"`
- `getAttribute`, `setAttribute`, `id`, `innerHTML` etc.

Look in the code and see how the game does it.

You have to cheat to win

How many ways can you cheat to win?

Examples of ways to cheat - challenges

- Amend Dom to show winning and show alert then take screenshot
- Amend the colour of the button to match the square with dev tools
- Amend the colour of the square to match the button with dev tools
- Amend the array to include "red" during a breakpoint debug mode
- Use Debugger to change running code variables

Examples of ways to cheat

- Use javascript in the console to
 - set square colour and press the button
 - set button colour to match square and press the button
 - set button and square to same colour
 - change the contents of the array
 - change the `checkColour` function
- If running from a server then use proxy to amend code prior to loading
 - Amend the code to remove the check or change it to `'if(true)'`

Demo Amending Dom and Debugging if necessary

Use JavaScript in the console to set square to red

```
document.getElementById("theSquare").  
    style.backgroundColor = "red";
```

Use JavaScript in the console to set button to grey

When the game starts, the square is `grey` so we could copy the default in the code and set the button to match.

```
document.getElementById("theButton").  
    style.backgroundColor = "grey";
```

Use JavaScript in the console to set button and square to same colour

```
document.getElementById("theSquare").  
    style.backgroundColor =  
        document.getElementById("theButton").  
            style.backgroundColor;
```

Use JavaScript in the Console to change the contents of the array

```
var colours = ["red", "red", "red", "red", "red"];
```

or even just

```
var colours = ["red"];
```

Use Console to amend the `checkColour` function

```
checkColour = function(){  
    changeStatusText("Winner!");  
    alert("You Win!");  
}
```

For bonus points:

- Change the colour of the square
- Change the link text to say "Change From red"

Bonus Points in checkColour function

```
checkColour = function(){  
    document.getElementById("theSquare").  
        style.backgroundColor = "red";  
    changeLinkText("Change From red")  
    changeStatusText("Winner!");  
    alert("You Win!");  
}
```

Questions?

The Coloured Square Changer Game

A variant of the Coloured Square Game.

- Marvel as it changes colour on its own
- Can you get a high score by matching the colours?

compendiumdev.co.uk/games/buggygames/the_coloured_square_game/colouredsquarechanger.html

Play/Test The Game

Aim - Get a High Score however you can without amending `score`

- What did you score?
- What did you notice?

Hints/Debrief

- Did you find the hidden link?
- Does `colourBlindModeEnabled` help?
- Does `debugMode` help?
- How can you take advantage of the missing `changeColour` function?

How Many Ways can you cheat?

Without amending `score` ?

Easy Examples

DOM

- enable and click on hidden `changeLink` link

JavaScript

- set `debugMode` variable to `true`
- set `colourBlindModeEnabled` variable to `true`
- reduce colours in array
- negative lives
- set timing variables to make game easier

Other Examples

- Infinite Lives Bot
- Re-enable `changeColour`
- Colour Change Bot
- Can you Create a Bot to Play the Game

debugMode and **colourBlindModeEnabled**

Give more information when playing. Which might help.

enable and click on hidden `changeLink` link

Use Dev tools to remove `display:none` then click on link to change colour when you want rather than wait on timer.

Reduce Colours in Array

```
colours = ["red"];
```

Makes the game a lot easier.

Negative Lives

The end of game check is for `lives===0` so if lives somehow goes -ve then the game never ends.

```
lives=-1;
```

Infinite Lives Bot

```
var infiniteLivesBot = setInterval(function(){  
    if(lives<2){lives=4;}  
},5);
```

```
clearInterval(infiniteLivesBot);
```

Re-enable `changeColour`

```
changeColour = function(){  
    document.getElementById("theSquare")  
        .style.backgroundColor=  
        document.getElementById("theButton").  
            style.backgroundColor;  
}
```

Colour Change Bot

```
var colourChangeBot = setInterval(function(){  
    var sq = document.getElementById("theSquare");  
    var bt = document.getElementById("theButton");  
  
    if(sq.style.backgroundColor!=  
    bt.style.backgroundColor){  
        sq.style.backgroundColor=  
        bt.style.backgroundColor;}  
},500);
```

```
clearInterval(colourChangeBot);
```

Create a Bot to Play the Game

```
var gamePlayBot = setInterval(function(){  
    var sq = document.getElementById("theSquare");  
    var bt = document.getElementById("theButton");  
  
    if(sq.style.backgroundColor===  
    bt.style.backgroundColor){  
        checkColour();}  
},100);
```

```
clearInterval(gamePlayBot);
```

To help it get started

```
timeToChangeSquare=500; timeToChangeButton=500;  
currentButtonChanges=10; currentSquareChanges=10;
```

Questions?

Bookmarklets

Any JavaScript code can be:

- wrapped in a function and called
- wrapped in a function and run immediately
- wrapped in a function and called as a url and stored as a bookmarklet.

e.g.

- worked example using create 100 todos at todomvc.com/examples/backbone

create 100 todos at console

```
for(x = 0; x < 100; x++) {  
  app.todos.create({  
    title: "todo ".concat(x),  
    order: app.todos.nextOrder(),  
    completed: false  
  })  
}
```

Same code in a function allocated to `t`

```
t = function(){  
    for(x = 0; x < 100; x++) {  
        app.todos.create({  
            title: "todo ".concat(x),  
            order: app.todos.nextOrder(),  
            completed: false  
        })  
    }  
}
```

run `t()` to create 100 todos

Anonymous Function - Runs Immediately

```
(function(){  
    for(x = 0; x < 100; x++) {  
        app.todos.create({  
            title: "todo ".concat(x),  
            order: app.todos.nextOrder(),  
            completed: false  
        })  
    }  
})();
```

As a Bookmarklet - run from URL

```
javascript:(function(){  
    for(x = 0; x < 100; x++) {  
        app.todos.create({  
            title: "todo ".concat(x),  
            order: app.todos.nextOrder(),  
            completed: false  
        })  
    }  
})();
```

Delete All Todos as a Bookmarklet

```
for(x = app.todos.models.length - 1; x >= 0; x--) {  
  app.todos.models[x].destroy()  
}
```

```
(function(){  
for(x = app.todos.models.length - 1; x >= 0; x--) {  
  app.todos.models[x].destroy()  
}})()
```

```
javascript:(function(){  
for(x = app.todos.models.length - 1; x >= 0; x--) {  
  app.todos.models[x].destroy()  
}})()
```

Questions?

Multi-User RestMud Game

RestMud is a Browser Based Text Adventure Game

A normal text adventure game would look like:

```
> look
You Look: There is an exit to the east
and a torch on the ground.
> take torch
You take the torch.
> go e
You go east. It is too dark to see.
> illuminate
You switch on the torch and can
see exits to the west and east.
```

A RestMud session would look like:

```
> GET /player/bob/look
You look
> GET /player/bob/take/torch
You took: torch. You now have the A Torch
> GET /player/bob/go/e
You Go East
> GET /player/bob/illuminate
Good work. You illuminated the 'A Torch
```

/user/verb/noun

Verbs:	Nouns
- illuminate, darken	
- go, open, close	e, w, n, s
- inventory	
- hoard, polish, use	<i>itemid</i>
- say	<i>something</i>
- take, drop	<i>itemid</i>
- help	

There may be other verbs

Hoard

When you see your 'hoard' in the location you can hoard treasure to score points e.g.

```
/player/bob/hoard/_treasureid_
```

Some treasures are 'junk' and score -ve points

- `/score`
- `/scores`

Polish things for more points when hoarding.

- `/inspect/_itemid_` to find out what it scores - but this costs points

Tips

- use dev tools and inspect the code
- make a map (the wiz might turn off the lights)
- hoard to score points

Demo of GUI

- start game
- use GUI to 'click' on verbs
- look at the URL
- inspect the DOM
 - look for `span` with IDs
- change URL directly when no link on screen

API

The 'Rest' in RestMud means it has a 'REST' interface.

- GET
- POST

API GET Requests

GET

- `http://_host_/api/player/_username_/_verb_/_noun_`

e.g.

```
GET http://localhost:4567/api/player/user/messages
```

API POST Requests

POST

- `http://host/api/player/username`
 - `body`
 - `{"verb": "inspect", "nounphrase": "cloth_2"}`
 - `{"verb": "look"}`

e.g.

```
POST http://localhost:4567/api/player/user
```

API Register

POST Register

- `http://host/api/register`
 - body
 - `{"username":"user", "displayname":"Mighty User", "password":"password"}`
 - if game requires a 'secret code'
 - add field `secret`

e.g.

```
POST http://localhost:4567/api/register
{"username":"user", "displayname":"Mighty User",
"password":"password", "secret":"iwejwjwjw"}
```

- session cookie automatically logged in for user after registering

API Register Response

```
{
  "status": "success",
  "data": {
    "X-RESTMUD-USER-AUTH":
      "be75b9a3-2b71-4b28-a92b-0295da34c814",
    "successMessage":
      "Registered user user as 'Mighty User'",
    "username": "user"
  }
}
```

- can add a custom header "X-RESTMUD-USER-AUTH" with GUID above as security to not require login

API Login

POST Login

- `http://host/api/login`
 - body
 - `{"username":"user", "password":"password"}`

e.g.

```
POST http://localhost:4567/api/login
```

API Login Response

```
{
  "status": "success",
  "data": {
    "resultoutput": {
      "lastactionstate": "success",
      "lastactionresult": "user bob2 is now logged in"
    },
    "userDetails": {
      "displayName": "Mighty User",
      "userName": "user",
      "authToken": "be75b9a3-2b71-4b28-a92b-0295da34c814"
    }
  }
}
```

- login response has a cookie that can be used for authentication

Message Authentication

- send the `JSESSIONID` cookie set on register or login or from GUI Session
- send the `X-RESTMUD-USER-AUTH` custom header in the HTTP request
- use basic auth with registered username and password

Recommended API Usage Tool

I use [postman](#) for most of my interactive REST API usage.

- <https://www.getpostman.com>

API Demo

- login
- look
- take stuff
- etc.

Multi-User RestMud Game Session

Register With Secret Code

<http://restmud.herokuapp.com/register>

Note: insecure site, do not use 'proper' username or password

Secret Code: ?????

More info:

<http://compendiumdev.co.uk/page/restmud>

RestMud and JavaScript

- RestMud URLs 'do' stuff
- and they are on the same domain as the web site
- we can hit them with JavaScript

XMLHttpRequest

XMLHttpRequest is JavaScript's way of making HTTP requests.

```
xmlhttp = new XMLHttpRequest();  
xmlhttp.open("GET", "/player/user/open/e", false);  
xmlhttp.send(null);
```

The above:

- creates an XMLHttpRequest object
- open a GET request on url /player/user/open/e (false means it is a synchronous request)
- send the request with a null body because it is a GET request

Example: Open all Doors

```
xmlhttp = new XMLHttpRequest();  
dirs = ["n", "s", "e", "w"];  
dirs.forEach(  
    function(dir){  
        xmlhttp.open("GET", "/player/user/open/"+dir, false);  
        xmlhttp.send(null);  
    }  
);
```

The above issues four `GET` requests to the server.

- `/player/user/open/n`
- `/player/user/open/s`
- `/player/user/open/e`
- `/player/user/open/w`

Why?

We could save that in a bookmarklet then we have options that are not available from the RestMud GUI.

- We could perhaps write a 'Take All' function.
- Or in the Inventory screen we could 'Hoard All'
- etc.

You can find more details on `XMLHttpRequest` online
w3schools.com/xml/ajax_xmlhttprequest_create.asp

Play

- 15 minutes
- Play either using GUI or API
- Highest score wins

Challenges

- Find at least 3 Bugs in the Grid Walker Game
- Get Highest Score in Number Hover Text Game
- Get Highest Score in RestMud
- Get the Highest Score on The Grid Walker
- Create a Bookmarklet to 'take all' in RestMud
- Create a Bookmarklet to 'close all doors' in RestMud

Single Player RESTMud

You can use the multi-user version of RESTMud:

- <http://restmud.heroku.app/register>

Or download the Single player version from:

- <http://compendiumdev.co.uk/page/atletstest2017>

Challenges

- Write a Bot to Play Number Hover Text Game
- Create a bot that plays and wins (does not cheat) The Random Walker
- Write a bot that wins (does not cheat) the One To Nine Calculator Game
- Create a bot that plays and wins (does not cheat) The Grid Walker
- Create a bot that plays and wins The Grid Walker (it is allowed to cheat once)

3rd Party Challenges

- Write an infinite supply of EMP bot for Z-Type <http://zty.pe/>
- Write a HUD for 3D Monster Maze
- Trap Rex in 3D Monster Maze

Make This ZType Bot Better: <http://zty.pe/>

- survive longer
- less noise
- more accurate

```
var zTypeBotAlphaChars = "abcdefghijklmnopqrstuvwxyz";
var zTypeBotCharToType = 0;
ztypebot = setInterval(function() {
    ig.game.shoot(
        zTypeBotAlphaChars.charAt(zTypeBotCharToType));
    zTypeBotCharToType = (zTypeBotCharToType + 1) % 26;
}, 100);
```

Remember to stop bot:

```
clearInterval(ztypebot);
```

Recap DOM and Dev Tools

- DOM Investigation with inspect
- Can Amend DOM
- Events link to code in Event Listener
- Code Exploration in Sources
- code snippets in Chrome

Recap JavaScript Capabilities

- Amend variables from console
- Run code from console
- `for` loops block processing
- use `setInterval` bots for automating in parallel
- bookmarklets to run code samples

Recap on practice apps

There are other games and apps listed here:

compendiumdev.co.uk/games/buggygames

See also reference section.

Questions?

Final Section

- You should be familiar with the dev tools now
- Learn to read HTML
- Learn to manipulate HTML with dev tools
- Learn to manipulate HTML with JavaScript

Why? Because... Testing

- Anything your app can do, you can do
- This is tactical automation
- You don't need to know much JavaScript to achieve a lot

Questions?

The End Of Workshop

- www.compendiumdev.co.uk

Reference Section

The following section is unlikely to be covered in the workshop.

It contains a reference on the JavaScript covered throughout the workshop - but in more detail.

Also Additional Games and Challenges.

Just Enough JavaScript To Be Dangerous

This section will provide a short reference to JavaScript. Not all of JavaScript you understand, just enough to be dangerous!

Console

Right click on your web page to access the developer tools.

Click on the 'Console' tab and you can start writing JavaScript.

Variables

```
var debugMode=true;
```

- The variable has a 'name' i.e. `debugMode`
- we could 'declare' it by missing out the `=true`
- we are initializing it (setting its value for the first time) with `=true`
- Add the ';' at the end of the line
- The variable has a `value` i.e. `true`
- we can type the variable name into the console to see information about it
- we assign it a new value with `=` e.g. `=false`

Variables can hold pretty much any type of information e.g.

- boolean : `true` , `false`
- String : `"Bob"`
- Integer : `'23'`
- functions
- objects

Operators

`--/+*`

- `++` to increment a variable
- `--` to decrement a variable
- `+=` to add a value e.g. `x+=5` (also `/=` , `*=` , `-=`)
- `'+'` can be used to add numbers and concatenate strings

Strings

Strings can be `""` or `' '`

- `substr(...)` - get sub string
 - `.substr(-2)` - the last 2 characters
 - `.substr(2)` - the string starting from the 2nd character
 - `.substr(2,8)` - the string from chars 2 to 8
- `split(...)` - create an array by splitting based on a char
 - `.split(",")`
- `parseInt("10")` - takes a string and returns an integer, whitespace is ignored, `NaN` returned if not a convertible String
 - developer.mozilla.org/en/docs/Web/JavaScript/Reference/Global_Objects/parseInt
- `.length` - length of string
- `.concat(...)` - return string with argument `...` concatenated

Integers

Hex

- `myNum.toString(16)` - convert to hex
- `var myNum = 0xFFFFFFFF;` - initialize a variable using HEX notation

Math library/Object

JavaScript has a standard Math library/Object to do math stuff

- `Math.random()`

w3schools.com/js/js_math.asp

Using the Console object

If I type `debugMode` into the console I will see the value.

But I can also print information to the console with `console.log`

e.g.

- `console.log('Hello');`
- `console.log(debugMode);`

Alerts and Prompts

We can make pop up dialogs appear.

```
alert("Hello");
```

```
prompt("type your name", 'no name');
```

- The second argument is optional
- prompt returns what the user types in

But this stops other JavaScript from running until the alert has been dismissed.

Note `""` and `''` are pretty much interchangeable.

Functions

Functions let us create code that we can call at any time we want.

```
function randomNumberUpTo(theVal){  
    return Math.floor(Math.random() * theVal)  
}  
randomNumberUpTo(10;)
```

- This is a `function`
- called `randomNumberUpTo`
- it takes a single argument `theVal`
- it uses the JavaScript `Math` library
- it returns a `random` number up to the value of `theVal`

Functions

We can allocate functions to variables.

```
var randomNum = function randomNumberUpTo(theVal){  
    return Math.floor(Math.random() * theVal)  
};  
randomNum(10);
```

Iteration

for Loops

```
for(x=0;x<100;x++){  
    console.log("line " + x);  
}
```

- loop from 0 to 99 ($x < 100$)
- increment by 1 each time ($x++$)
- each time run the code within `{}`
 - log to the console `"line "` with `x` appended e.g. 'line 0', 'line 1'

Arrays - Position Index

```
var colours = ["red", "green", "blue", "grey", "black"];
```

- An array that is indexed by position
 - `colours[0]=== "red"` arrays start at index 0
 - `colours[5]===undefined` last index in the array is 4
- Arrays have length
 - `colours.length===5`
- The last index in an array is `length-1`
 - `colours[colours.length-1]=== "black"`
- We can set the values in the array
 - `colours[0] = "white";` the first item is now "white"
- `.join(",")` - create a string by concatenating all elements together with `","`

Arrays - Key Indexed

Arrays can be 'key' indexed;

```
var colours = [];  
colours["stop"]="red";
```

Then `colours["stop"]==="red"` is `true`.

Arrays vs Objects

Arrays and Objects are very similar

```
var colours = {"stop": "red"};
```

Then `colours["stop"] === "red"` is also `true`.

- Objects `{}`
- Arrays `[]`

Selection and Conditions - `if` statements

```
var debugMode=true;
if(debugMode===true){
    console.log("debug mode is on");
}else{
    console.log("debug mode is not on");
}
```

JavaScript has strange equality operators:

- `===` check if something equals

Run Code Every X milliseconds

`setInterval` runs a function every 'X' milliseconds

```
var sloganbot = setInterval(changeSlogan,1000)
```

- every 1000 milliseconds run the function `changeSlogan`

```
var sloganbot = setInterval(function(){changeSlogan()},1000)
```

- every 1000 milliseconds run the 'anonymous' function
 - in this case the 'anonymous' function calls the `changeSlogan` function
- You can add as much code as you want in an 'anonymous' function

Stop an 'interval' with 'clearInterval'

```
clearInterval(sloganbot)
```

Interact with the DOM

Finding Elements

- `getElementById`
- `getElementsByClassName`
- `getElementsByName`
- `getElementsByTagName`
- `querySelector`
 - use CSS selector as the query argument to match first element
- `querySelectorAll`
 - return all in an array using CSS selector

can find from an element or `document`

Change Style Element Attributes

```
document.getElementById("theSquare").  
    style.backgroundColor = "red";
```

- get individual elements with `getElementById`
- access the `style` attribute of an element with `.style`

Hiding an element:

```
document.getElementById("theSquare").style.display="none"
```

- set to "block" or "" to display it

Change Element Attributes

```
document.getElementsByClassName("title")[0].  
    setAttribute("style", "display:none");
```

```
document.getElementsByClassName("title")[0].  
    getAttribute("style");
```

Change Element Text

```
elem.innerHTML = "Hello";
```

Changing Event Code

The system might have defined events.

```
<div class="title">  
  <button class="title" onclick="changeSlogan()">  
    I Want A Slogan  
  </button>  
</div>
```

We can change them.

Changing Event Code

```
onclick="changeSlogan()"
```

```
var theButton=document.getElementsByClassName("title")[2];  
theButton.onclick = function() {  
    alert('hello');  
};
```

Changing Event Code with Listeners

```
var theButton=document.getElementsByClassName("title")[2];
theButton.onclick = null;
theButton.addEventListener("click", function(){
    alert('from adding event');
}, false);
```

- The `onclick` attribute will operate alongside event handlers.
- `false` is for 'useCapture' - true for capturing phase, false (default) for bubbling phase
 - bubbling for handling child events
 - capturing - has to be element clicked on
- 'useCapture' is optional

Removing Event Code Listeners

Create function into a variable if we want to remove the event Listener

```
var theButton=document.getElementsByClassName("title")[2];
theButton.onclick = null;
var eventHandlingFunction =
    function(){alert('from adding event');}
theButton.addEventListener("click",
    eventHandlingFunction, false);
```

Same phase must be specified to remove it.

```
theButton.removeEventListener("click",
    eventHandlingFunction, false);
```

Bookmarklets

Any JavaScript code can be:

- wrapped in a function and called
- wrapped in a function and run immediately
- wrapped in a function and called as a url and stored as a bookmarklet.

e.g.

- worked example

Build up code for bookmarklet

write out 100 lines to console

```
for(x = 0; x < 100; x++) {console.log(x)}
```

Same code in a function allocated to `t`

```
t = function(){  
  for(x = 0; x < 100; x++) {console.log(x)}  
}
```

run `t()` to write 100 lines

Anonymous Function - Bookmarklet

Anonymous Function - Runs Immediately

```
(function(){  
    for(x = 0; x < 100; x++) {console.log(x)}  
})();
```

As a Bookmarklet - run from URL

```
javascript:(function(){  
    for(x = 0; x < 100; x++) {console.log(x)}  
})();
```

Bookmarklet Resources From Alan

- a Tool
 - <http://compendiumdev.co.uk/apps/eviltester/bookmarklets.html>
 - <https://github.com/eviltester/bookmarklet>
- Blog Post
 - <http://blog.eviltester.com/2017/04/a-quick-intro-to-bookmarklets.html>

Bookmarklet Resources

Resources:

- <http://adrianroselli.com/2015/01/css-bookmarklets-for-testing-and-fixing.html>
- <http://www.dev-hq.net/posts/1--create-javascript-bookmarklet>
- <http://www.squarefree.com/userstyles/make-bookmarklet.html>
- <https://www.smashingmagazine.com/2010/05/make-your-own-bookmarklets-with-jquery/>

XMLHttpRequest

XMLHttpRequest is JavaScript's way of making HTTP requests.

```
xmlhttp = new XMLHttpRequest();  
xmlhttp.open("GET", "/player/user/open/e", false);  
xmlhttp.send(null);
```

The above:

- creates an XMLHttpRequest object
- open a GET request on url /player/user/open/e (false means it is a synchronous request)
- send the request with a null body because it is a GET request

More about XMLHttpRequest

You can find more details on XMLHttpRequest online
w3schools.com/xml/ajax_xmlhttprequest_create.asp

This is very handy if your API is on the same domain as the web site and there are no cross domain issues.

Also if you know the URL structure of an application then you can hit any Forms or GET requests (if they have side-effects).

Good JavaScript References For Future Study

Books

- [Eloquent JavaScript](#)
- [JavaScript for Cats](#)

Tutorials

- [\[w3schools.com/js\]http://www.w3schools.com/js/](http://www.w3schools.com/js/)
- [Mozilla Developer Network Javascript](#)
- html5dog.com/guides/javascript
- learn-javascript.org

Good JavaScript References For Future Study

References

- w3schools.com/jsref

Links

- [Essential JavaScript Links](#)

EvilTester Sloganizer Application

A single page application. That generates random slogans related to software testing.

It may well have bugs.

- Live Version EvilTester.com/apps/sloganizer.html
- Version 1
EvilTester.com/apps/sloganizer/version/1/sloganizer.html
- Source github.com/eviltester/sloganizer

Visit Experiment and Test

- Can you find any bugs in version 1?

- Version 1

EvilTester.com/apps/sloganizer/version/1/sloganizer.html

- Look at the Code
- What can you make it do?
- Any differences between early version and live version?
 - Live Version EvilTester.com/apps/sloganizer.html

Debrief

- What did we learn?

From The Code

Main Entry Point

```
<div class="title">  
  <button class="title" onclick="changeSlogan()">  
    I Want A Slogan  
  </button>  
</div>
```

Top Level Objects:

- phrases
- sentences

Exercises

- Look at the source for the sloganizer and see how much you understand
- type the names of the objects that you see into the console
- explore the output in the console
- run some of the functions that you find
- Generate a random sentence using a `sentences` template
 - `process_sentence` with an item from the `sentences` array
- Generate 100 random sentences from a specific template
- Generate 1000 random sentences
- Generate a new slogan in the GUI every second
- Hack the app to spit out your own slogans

Generate a random sentence using a sentences template

```
process_sentence(sentences[0]);
```

Can you see the difference between these approaches?

```
console.log(process_sentence(sentences[0]));
```

Generate 100 random sentences from a specific template

```
for(x=0;x<100;x++){  
    console.log(process_sentence(sentences[0]));  
}
```

Generate 1000 random sentences

```
for(x=0;x<1000;x++){  
    console.log(random_sentence());  
}
```

Generate a new slogan in the GUI every second

```
var sloganbot = setInterval(function(){changeSlogan()},1000)
```

```
clearInterval(sloganbot)
```

Hack the app to spit out your own slogans - using existing structure

- `sentences[8]` is simple `"#o_just #just_because"`

```
phrases['o_just']=["Since when did I",  
                  "But you always", "He will rudely"];  
phrases['just_because'] = ["kick a football",  
                           "smoke bananas", "sing on the phone"];
```

```
for(x=0;x<100;x++){  
    console.log(process_sentence(sentences[8]));  
}
```

Hack the app to spit out your own slogans - using new structure

Design:

- "I like eating mice"
- sentence: "#i_do_like #i_do_like_doing #i_like_doing_with"
- i_do_like: "I like", "I hate", "I despise", "I loathe", "I love"
- i_do_like_doing: "eating", "smashing", "feeding", "teasing", "tickling", "destroying", "making", "teaching"
- i_like_doing_with: "cats", "dogs", "mice", "peers", "Systems", "Managers", "Testers", "Developers", "code", "beliefs", "memories", "attitudes"

Hack the app to spit out your own slogans - using new structure

Implementation:

```
sentences.push(
    "#i_do_like #i_do_like_doing #i_like_doing_with");
phrases['i_do_like']=
    ["I like", "I hate", "I despise", "I loathe", "I love"];
phrases['i_do_like_doing']=
    ["eating", "smashing", "feeding", "teasing",
    "tickling", "destroying", "making", "teaching"];
phrases['i_like_doing_with']=
    ["cats", "dogs", "mice", "peers", "Systems",
    "Managers", "Testers", "Developers", "code",
    "beliefs", "memories", "attitudes"];
```

```
for(x=0;x<100;x++){
    console.log(process_sentence(sentences[sentences.length-1]));
}
```

I T.E.S.T With A.T.T.I.T.U.D.E app

A T-Shirt slogan generating app.

```
I  
T.E.S.T  
with  
A.T.T.I.T.U.D.E
```

Where the acronyms would expand out to:

- T.E.S.T - Tactical, Exploratory, Strategic, Testing
- A.T.T.I.T.U.D.E - Attack, Tenacious, Task, Interrogate, Technical, Understand, Debrief, Experiment
- eviltester.com/apps/testwith/version/1/testwith.html

Exercise

- visit the application page
 - eviltester.com/apps/testwith/version/1/testwith.html
- Explore it and see if you can identify what functionality is available
 - Inspect the DOM
 - look in the source
 - look at the comments

Treat the investigation of the application itself as a 'game'.

Debrief

- What did we learn?
- What did you see?
- What did you make it do?
- Issues?

Exercise: Investigation Hints

- Type in text, then? Clear?
- Inspect
 - `div` (s) with `id` (s)
- `[clear]` calls `makeExplodedSentenceForm` what does that do?
- non-visible `gtrig` - why? what does it do?
- what uses the canvas?

Debrief

- What did we learn?
- What did you see?
- What did you make it do?
- Issues?

More About?

- [Render] button
- [Header] button

Keep Going

ZType Bots

These bots run against zty.pe

Crude Bot

```
var zTypeBotAlphaChars = "abcdefghijklmnopqrstuvwxyz";
var zTypeBotCharToType = 0;
ztypebot = setInterval(function() {
    ig.game.shoot(
        zTypeBotAlphaChars.charAt(zTypeBotCharToType));
    zTypeBotCharToType = (zTypeBotCharToType + 1) % 26;
}, 100);
```

Remember to stop bot:

```
clearInterval(ztypebot);
```

Good Bot

A Z-Type Bot

```
ztypebot = setInterval(function() {  
    if (ig.game.currentTarget == null) {  
        for (var t in ig.game.targets) {  
            if (ig.game.targets[t].length > 0) {  
                ig.game.shoot(t);  
                break;  
            }  
        }  
    }  
  
    } else {  
        while(ig.game.currentTarget != null){  
            ig.game.shoot(  
                ig.game.currentTarget.remainingWord.charAt(0));  
        }  
    }  
}, 2);
```

BIO - Alan Richardson

Alan Richardson has more than twenty years of professional IT experience, working as a programmer and at every level of the testing hierarchy from tester through head of testing. He has performed [keynote speeches and tutorials at conferences](#) worldwide. Author of [multiple books on testing and automating](#). Alan also has created [online training courses](#) to help people learn [Technical Web Testing](#) and [Selenium WebDriver with Java](#). He works as an [independent consultant](#), helping companies improve their use of automation, agile, and exploratory technical testing.

Alan Richardson (www)

- www.compendiumdev.co.uk
- www.eviltester.com
- www.seleniumsimplified.com
- www.javafortesters.com
- uk.linkedin.com/in/eviltester
- testerhq.com - Aggregated Feed

Follow

- LinkedIn - [@eviltester](#)
- Twitter - [@eviltester](#)
- Instagram - [@eviltester](#)
- Facebook - [@eviltester](#)
- Youtube - [EvilTesterVideos](#)
- Pinterest - [@eviltester](#)
- Github - [@eviltester](#)
- Slideshare - [@eviltester](#)

Books Written By Alan Richardson

- Java For Testers
- Dear Evil Tester
- Automating and Testing a REST API
- Selenium Simplified

Online Training Courses By Alan Richardson

- Selenium Webdriver With Java
- Technical Web Testing 101
- Evil Tester Talks: Technical Testing
- Case Study: Java Desktop Application Technical Training

The End